

# Smart-Pikachu: Extending Interactivity of Stuffed Animals with Large Language Models

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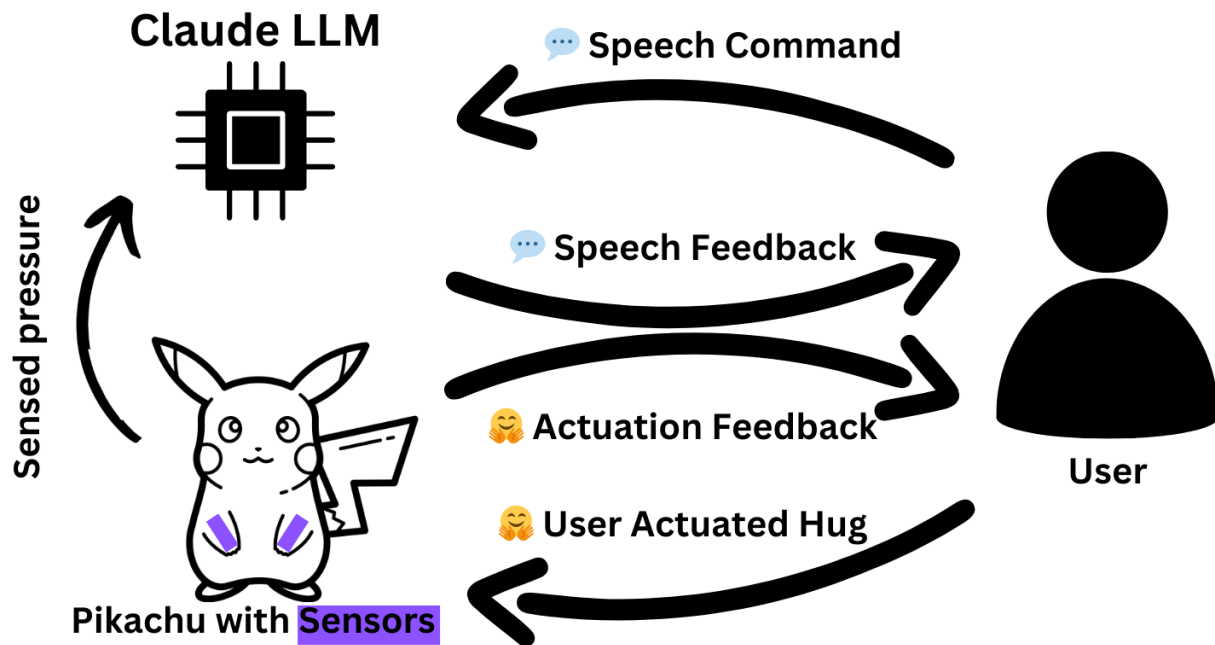


Figure 1: Overview of Pikachu’s interaction and feedback loops.

## ABSTRACT

We propose Smart-Pikachu, a stuffed animal equipped with sensing and actuation to explore the use of large language models (LLM’s) with sensor data inputs. The augmentation of pressure sensing will allow for the LLM to interpret various interactions such as hugs and handshakes with the user. Furthermore, the actuation capabilities will extend our system’s interactivity by providing physical feedback to the user. We will also incorporate text-to-speech output from the LLM to add another mode of interaction between the system and user. In this Student Innovation Challenge,

we intend to explore applications at the intersection of sensing and interaction through LLM’s and demonstrate an extension of LLMs’ multimodal capabilities.

## CCS CONCEPTS

• **Human-centered computing** → *Interaction devices*; • **LLM**; • **Sensing**; • **Interaction**;

## KEYWORDS

Empathetic Computing, Prompt Design

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## 1 INTRODUCTION

Large language models (LLM's) have gained significant traction in the last year for a wide variety of tasks. They are already being used for productivity and automation, research and learning, and creativity and play. For example, LLM's now power and augment search engine results, a previously plateaued domain. LLM's are also revitalizing education, enabling chatbots to become effective, personalized tutors. These models develop their capabilities through training on massive text datasets, up to and including the entire Internet.

Despite their all of their sophistication, LLM agents are limited in their expressivity. As the name suggests, LLM's are fundamentally designed to process and generate textual language. More recent work has explored techniques for adapting LLM's to work with other types of data, such as GPT-4's ability to accept image inputs [2]. Other recent multi-modal LLM work has proposed the use of LLM's for training a student model for embodied agents [6]. Building on this related work, we are interested in investigating how these advances might enhance a physical agents' ability to interact with humans.

We propose Smart-Pikachu, a stuffed animal augmented with fabric pressure sensors and mechanical actuators. Pikachu's pressure sensors will enable it to recognize different kinds of interaction, such as hand shakes and hugs. In turn, it will also be able to respond with its mechanical actuators. For example, they will enable Pikachu to hug the user back, or hold the user's hand with its paw.

In this UIST SIC proposal, we will explore the use of LLM's for creating a Specifically, we propose the following plan:

- (1) Augment our Pikachu stuffed animal with pressure sensors and compliant actuators
- (2) Investigate how to design prompts that help Claude process the pressure sensor signals and generate corresponding responses through its mechanical actuators
- (3) Illustrate a design space in which we explore applications for a multi-modal interactive Pikachu

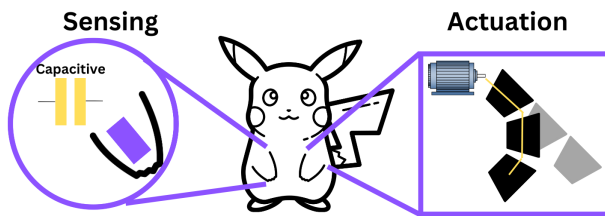


Figure 2: Sensing and actuation in Pikachu system.

## 2 PIKACHU SYSTEM

Our proposal manifests itself in the form of a Pikachu, a stuffed animal. We will augment an off-the-shelf Pikachu toy with pressure sensing and actuation capabilities.

We will employ pressure sensing capabilities similar to the fabric-based capacitive sensors introduced in FabToy [4]. These sensors will enable the system to recognize different types of interactions

such as hand-shakes and hugs. The sensor signals will be processed by Claude to recognize interactions similar to the techniques introduced in [5].

Moreover, we will adopt techniques previously used for actuating robotic grippers [1, 3] to power Pikachu's arms. These actuators will allow our system to further provide physical interactions such as reciprocating to hugs and handshakes. These actuators will allow our system to further provide physical interactions, such as reciprocating hugs and handshakes. We will devise techniques for processing Claude's output and mapping it to actuation patterns.

In addition to Pikachu's pressure sensing and actuation, we will also include a speaker and microphone system for speech input and output. We hope that Pikachu's multi-modality creates a richer, more engaging interaction experience.

## 3 ETHICS AND BROADER IMPACT

In order to ensure that no users are harmed with the mechanical actuators, the finger mechanism will be built with a limited range of motion. The firmware to control the servo motor attached to the actuating finger will also be constraint.

During the demo, users will be given the option to interact with Pikachu with or without actuation.

We believe this work will introduce an approach to improve the variety of input modalities into LLMs. Furthermore, we hope that our demonstration can bring more awareness towards the utility of timeseries sensing in the realm of LLMs.

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